

LEVERAGING ARTIFICIAL INTELLIGENCE IN DRONE DELIVERY SYSTEMS

LCE 116
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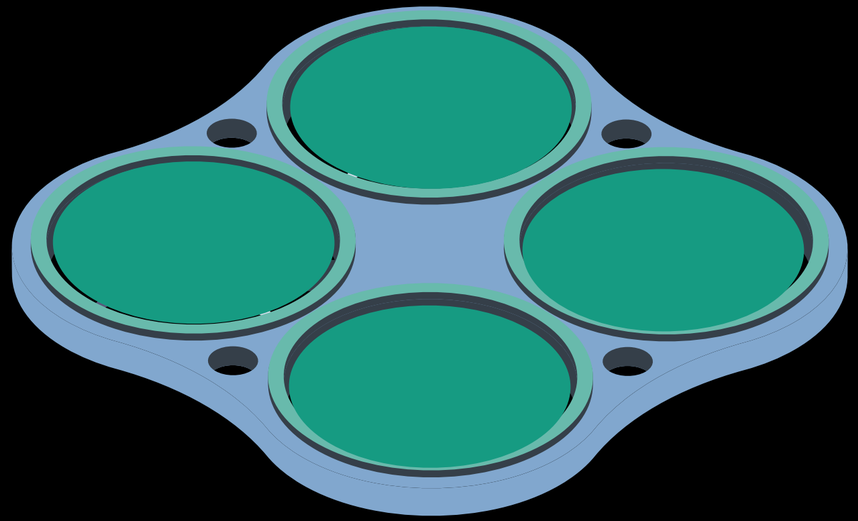
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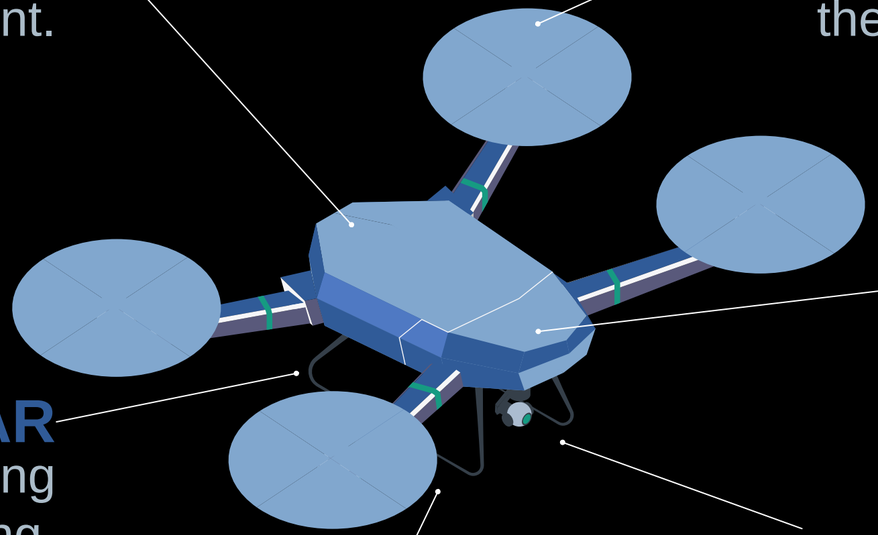
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INTRODUCTION



PARTS OF A DRONE



MOTOR
Powers the propellers and controls movement.

PROPELLER
Spins to lift and move the drone.

RECEPTOR
Receives signals and sends them to the AI for decisions.

LANDING GEAR
Protects the drone during takeoff and landing.

CAMERA
Used by the AI to see and make flight decisions.

DELIVERY PACKAGE
The Package that the Drone Delivers to the Client.

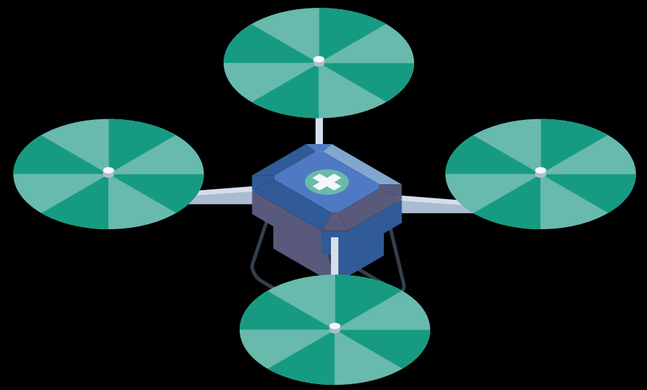
- **2.14B online shoppers in 2021** → Demand for faster deliveries
- Businesses race to meet soaring expectations for deliveries

PURPOSE

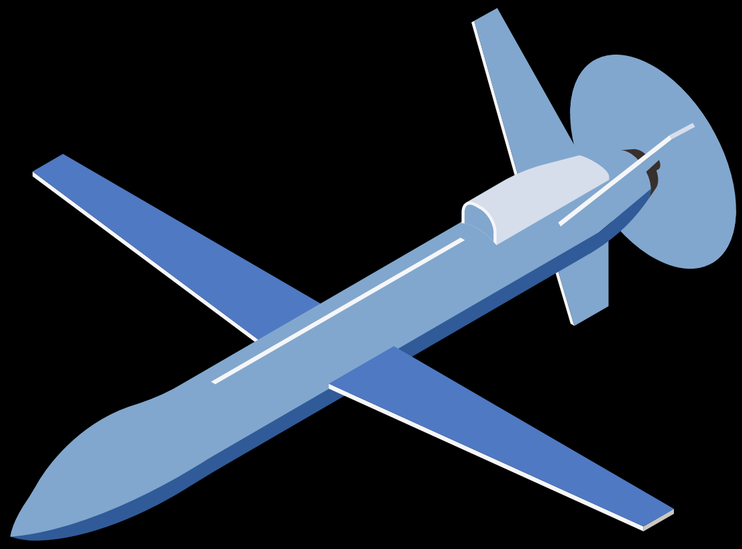
- Cut costs, speed last-mile delivery with AI-driven drones
- Key AI features: navigation, real-time routing, predictive analytics

RELEVANCE

- Gain a competitive edge & attract tech-savvy customers
- Boost customer satisfaction with faster, reliable deliveries
- Promote innovation & sustainability in modern business



02



WHAT ARE AI-DRIVEN DRONE DELIVERIES?

DEFINITION

- **Drones (unmanned aerial vehicles) equipped with AI for autonomous or semi-autonomous flight, route optimization, and decision-making.**

Computer Vision:

- Uses cameras & sensors to detect obstacles, read terrain, and ensure safe navigation.

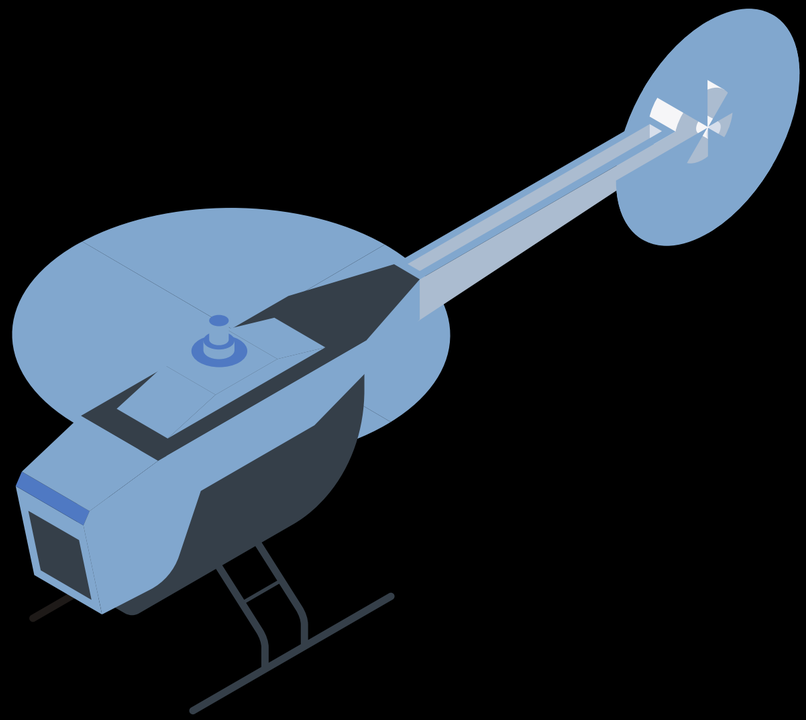
Machine Learning & Deep Learning:

- Real-time data processing for route optimization, obstacle avoidance, and predictive maintenance.

Edge Computing

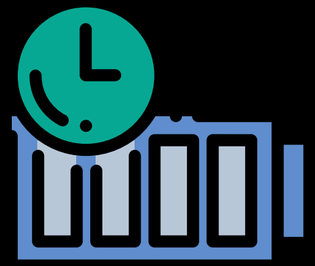
- Onboard processing to reduce latency and make faster decisions mid-flight.

03



BUSINESS APPLICATIONS & BENEFITS

Last-Mile Delivery Optimization



- Cuts delivery time with AI-optimized routes (avoids traffic)
- Reaches remote locations quickly for on-time delivery

Supply Chain & Logistics



- Automates internal logistics (e.g. warehouse transfers)
- Streamlines inventory movement & fulfillment
- Lowers costs (fuel, labor)

Data-Driven Decision Making

- Collects real-time delivery data
- AI insights for route optimization & resource allocation
- Enhances forecasting & continuous improvements



Competitive Advantages

- Distinguishes businesses with cutting-edge service
- Faster deliveries → higher customer satisfaction
- First-mover advantage & innovative brand image

04



CHALLENGES & CONSIDERATIONS



Regulations & Airspace Control

- Varying rules on where and how drones can fly
- Compliance with aviation authorities is crucial

Infrastructure & Tech Gaps

- Need secure landing hubs, recharging stations, or “drone lanes”
- AI relies on high-quality data and stable communications



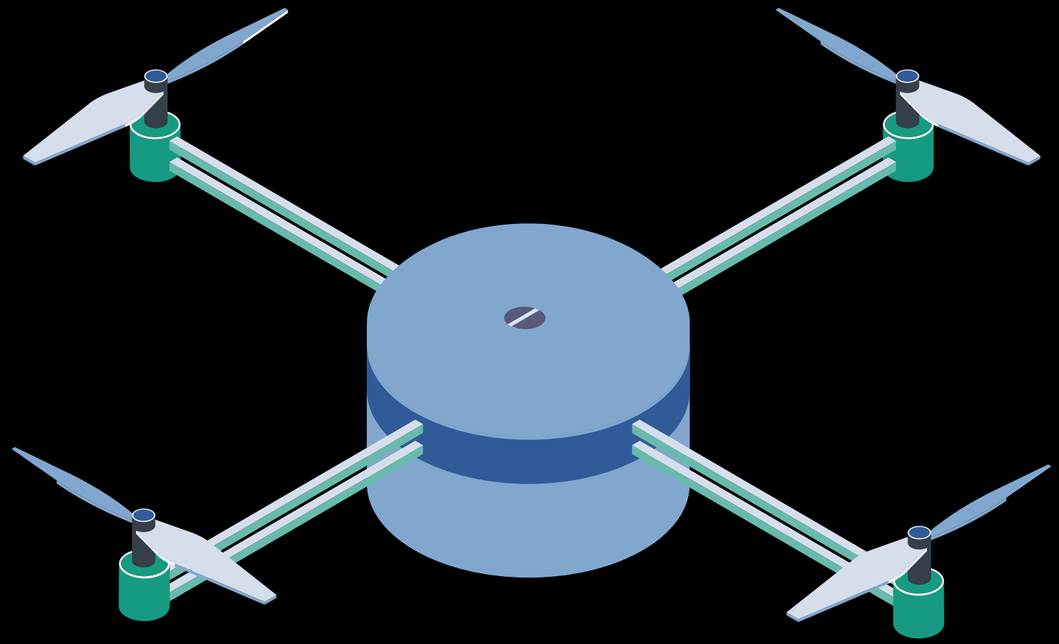
Security & Privacy

- Potential hacking or misuse of drones
- Privacy concerns over flights near private property

Public Acceptance

- Noise, safety worries, and public trust factor into success
- Address local concerns to gain community support

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CONCLUSION

Summary

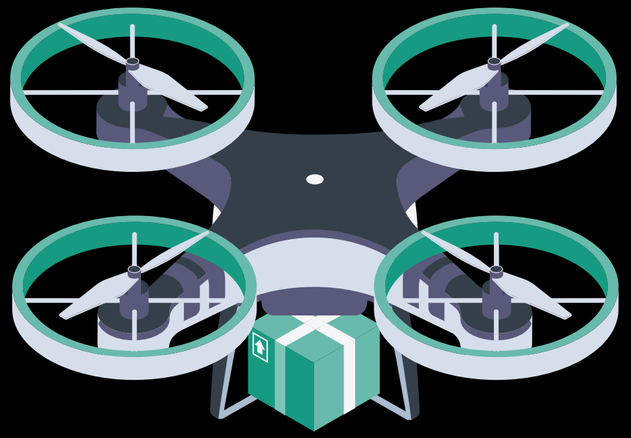
- AI-driven drone delivery can drastically increase speed, reduce costs, and boost competitive advantage.

Looking Ahead

- As technology matures and regulations adapt, drones could become standard in supply chain operations.

Call to Action

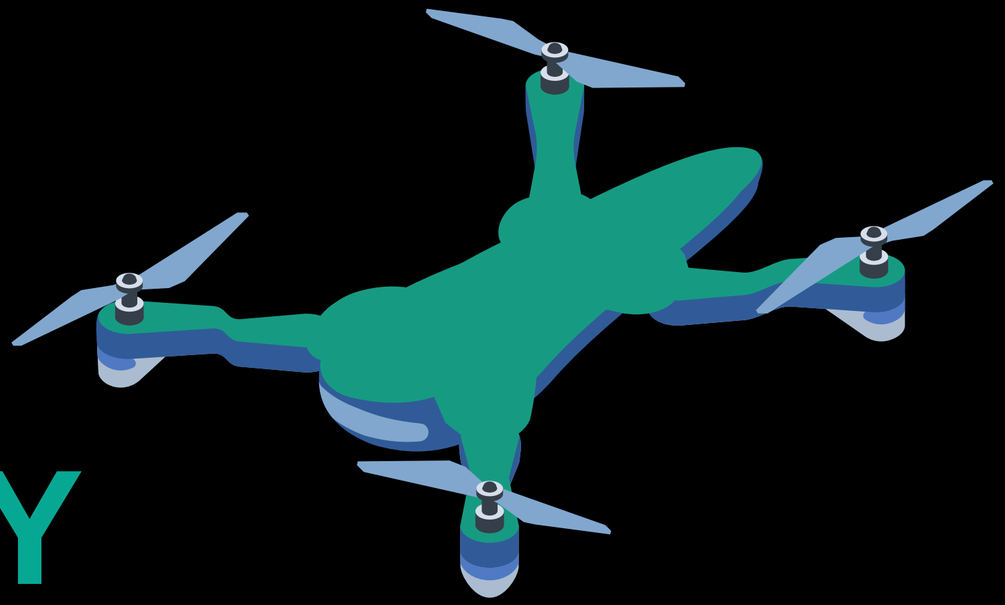
- Urge business leaders, policymakers, and customers to invest in research, infrastructure, and pilot programs to refine drone technology and expand its benefits.





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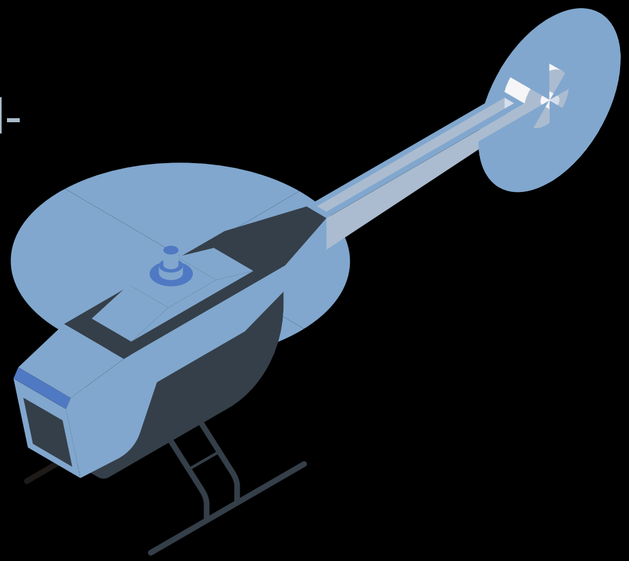


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THANK YOU!

DO YOU HAVE ANY QUESTIONS?

